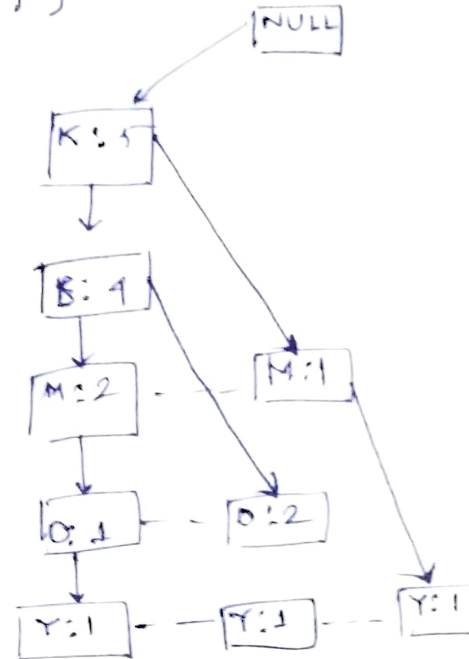


Frequent Pattern Growth Algorithm

conditional Pattern Base:

Try

<u>Item</u>	<u>conditional Pattern Base</u>
Y	$\{\{K, E, M, O:1\}, \{K, \cancel{E}, O:1\}, \{K, M:1\}\}$
O	$\{\{K, E, M:1\}, \{K, E:2\}\}$
M	$\{\{K, E:2\}, \{K:1\}\}$
K	$\{K:4\}$



Now, for each item the conditional pattern test is computed which is path levels of all the path which lead to any node of a given item in the frequent pattern tree. Now

Now for each item the conditional frequent pattern tree is built. It is done by taking a set of elements which is common in all of paths in the conditional pattern base of that item & calculating it's support count by summing the support count by all the path in the conditional pattern base

conditional frequent pattern tree

~~{K:4}~~

Items	conditional pattern tree	Pattern tree
Y	$\{\{K, E, M, O:1\}, \{K, E, O:1\}, \{K, M:1\}\}$	$\{K:3\}$
O	$\{\{K, E, M:1\}, \{K, E:2\}\}$	$\{K, E:3\}$
M	$\{\{K, E:2\}, \{K:1\}\}$	$\{K:3\}$
E	$\{K:4\}$	$\{K:4\}$
K		

CPB $\rightarrow$ conditional pattern Base

Now, for each item the conditional frequent pattern tree is built. It is done by taking the set of elements which is ~~same~~ common in all the paths in the CPB. of that item and calculating its support count by summing the support counts of all the paths in CPB.

Items	frequent pattern Generator
Y	$\{\langle K, Y:3 \rangle\}$
O	$\{\langle K, O:3 \rangle, \langle E, O:3 \rangle, \langle E, K, O:3 \rangle\}$
M	$\{\langle K, M:3 \rangle\}$
E	$\{\langle E, K:3 \rangle\}$
K	$\{\langle E, K:3 \rangle\}$

from this table we can able to make some different associations rules of item set of 2 & 3.

CLARA

CLARA Algo:

- 1) Instead of taking the whole set of data into consideration, a small portion of actual data is chosen as a representative ~~set~~ of data.
- 2) Medoids are being chosen from this sample ~~using~~ using PAM.
- 3) Sample is selected randomly.
- 4) ~~CLARA~~ CLARA draws multiple samples of the dataset, applies PAM of each sample & return it's base ~~cluster~~ clustering as an output.
- 5) The effective base of CLARA depends ~~set~~ on the sample size.
- 6) Out of all the sample whose cost is less select that one.
- 7) It is computationally fast.

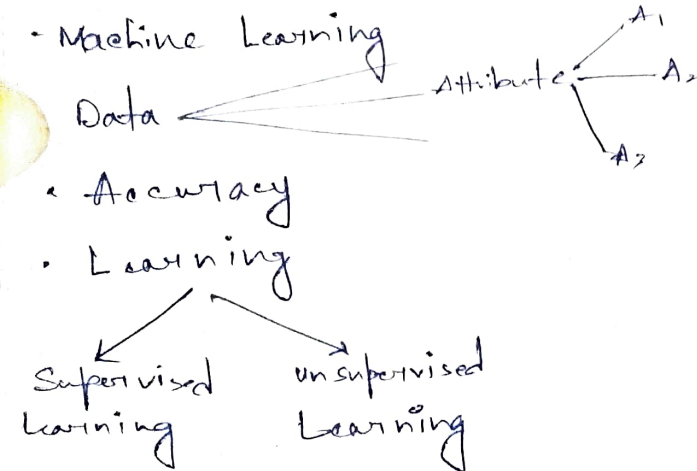
CLARAN Algo:

- 1) CLARAN algo thinks that the solution as a graph problem where each node is combination of K medoids.
- 2) If there are 10 points & $K=2$, No of possible node is ${}^{10}C_2$ ${}^{10}C_2$. & the nodes are represented by $P_1 \times P_2$ which are any 2 points of the table.
- 3) Each node is associated with the cost based on similarity ~~with~~ dissimilarity ~~set~~ with all other objects.
- 4) So ~~PAM~~ PAM is searched for a base node with minimum cost from this graph. As no. of nodes very high it is computationally very expensive.

Limitation of CLARA =

1. the best K medoids may not be selected during the sampling process. In this case CLARA will never find the best clustering.
2. If the sampling is biased or partial we cannot find good quality clusters.

Learning Mechanism:



Knowledge acquired through studying, experience, or being taught is called learning

What is ML

subfield of CS that evolved from ML is subfield of CS that evolved from the study of pattern recognition & computation learning. ML explores the study & construction of algo that can learn from and prediction of data such as algo operate by building a model for example - inputs in order to make data driven predictions or decisions.

system is evaluated by task performance.
However, a reinforcement system is said to have
learned to perform the task if after
learning the system's performance or J
increases as measured by an external reward.
The basic model of the system to perform
a task is compared to a reinforcement.